

NURSERY LINK UGANDA

A Web-GIS Marketplace and Supply-Gap Platform for Uganda's Tree Nursery Sector

SOFTWARE REQUIREMENTS SPECIFICATION DOCUMENT

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APPROVALS

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List of Acronyms and Abbreviations

API: Application Programming Interface

GIS: Geographic Information System

GPS: Global Positioning System

GUI: Graphical User Interface

MoMo: Mobile Money

MWE: Ministry of Water and Environment

NFA: National Forestry Authority

NGO: Non-Governmental Organisation

PostGIS: Spatial database extension for PostgreSQL

REST: Representational State Transfer

SMS: Short Message Service

SRS: Software Requirements Specification

UI: User Interface

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of Nursery Link Uganda, a Web-GIS marketplace and supply-gap analysis platform for Uganda's tree nursery and seedling supply sector. This document serves as a comprehensive guide for the system's stakeholders, including the development team, restoration planners, nursery operators, project supervisors, and prospective end-users, by outlining the functional and non-functional requirements, constraints, assumptions, and dependencies necessary to deliver a reliable and trustworthy platform.

This SRS also establishes a shared understanding among stakeholders about the goals, scope, and functionality of Nursery Link Uganda, and serves as a foundation for the system's design, development, testing, and deployment phases.

1.2 Document Conventions

- i. Bold text indicates section headers and important terms.
- ii. Italicized text is used for emphasis.
- iii. Numbered lists represent ordered sequences or steps.
- iv. Bulleted lists represent unordered items.

1.3 Intended Audience

- i. Project Supervisors and Reviewers: for planning, oversight, and evaluation.
- ii. Software Developers: for understanding system requirements to guide design and implementation.
- iii. Quality Assurance: for developing test cases and validation procedures.
- iv. Nursery Operators, Restoration Planners, NGOs, and Government Partners: for understanding system capabilities relevant to their role.
- v. Farmers and Buyers: as the primary consumers of the platform's discovery and ordering functions.

1.4 Scope

Nursery Link Uganda unifies Uganda's fragmented tree seedling supply chain by bridging the communication gap between commercial and community nurseries, ecosystem restoration planners, and individual smallholder farmers. The platform incorporates five core modules: Home, Nurseries (the Web-GIS discovery and marketplace engine), Free Seedlings (a subsidy directory), Library (a species encyclopedia), and News. An administrator-facing Nursery Shadow analytics layer overlays nursery service zones against satellite-derived forest-loss data to identify underserved, heavily deforested areas, directly extending the gap-analysis methodology developed for Mukono District in the underlying research concept note to the wider platform.

1.5 References

- i. Draft Concept Note for Developing a Web-GIS Platform for Tree Nurseries and Finding Seedling Supply Gaps in Mukono District, Uganda.
- ii. Nursery Link Uganda design blueprint (platform navigation and component specification).

- iii. Hansen et al. (2013). Global Forest Change dataset.
- iv. Data Protection and Privacy Act, 2019 (Uganda).
- v. IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.

2. Overall Description

2.1 Product Perspective

Nursery Link Uganda is a public-facing Web-GIS platform that unifies nursery discovery, seedling ordering, subsidy access, and species research into a single interactive map-driven experience. It replaces the static, non-spatial PDF registries and paper logbooks currently used by local forestry offices with a live, coordinate-accurate digital atlas. The platform interfaces with:

- i. A spatial database (PostgreSQL with the PostGIS extension) to store nursery locations, administrative boundaries, and inventory records.
- ii. A Ugandan mobile money API (MTN MoMo / Airtel Money) to process seedling payments and hold funds in escrow.
- iii. An SMS gateway to notify nursery operators of confirmed orders.
- iv. ArcGIS Survey123 for field data collection during nursery mapping campaigns.
- v. A satellite-derived forest-change dataset (the Global Forest Change dataset) to support Nursery Shadow analytics.

2.2 Product Functions

2.2.1 Account Management and Authentication

Visitors may browse nursery listings, species information, and news without an account. Authentication is required to place an order, apply to a free seedling campaign, or manage a nursery's inventory as an operator.

2.2.2 Nursery Discovery and Web-GIS Mapping

The Nurseries module presents a split-screen workspace: an interactive map engine on one side and a smart search and filter column on the other, allowing users to browse nurseries at national, district, and sub-county levels.

2.2.3 Nursery Marketplace and Order Management

Buyers can view a structured nursery profile card, choose to be routed to the nursery for self-pickup, or check out species and quantities for delivery, with automatic distance-based pricing and mobile money payment.

2.2.4 Free Seedlings (Gifts) Module

An equity-focused directory of subsidized seedling campaigns funded by government, NGOs, and corporate sponsors, filterable by sub-county and restoration purpose, with eligibility checklists to protect stock from commercial resale.

2.2.5 Digital Tree Library

A public species encyclopedia organised by functional category (Indigenous, Agroforestry, Exotic, Ornamental, and Medicinal), giving users the botanical detail needed to choose the right species before ordering.

2.2.6 News and Sector Advisories

A category-filterable feed of weather advisories, market alerts, and National Forestry Authority policy and grant announcements.

2.2.7 Nursery Shadow Analytics (Administrator)

An internal analytics layer that overlays road-network nursery service zones against satellite-derived forest-loss data to flag underserved, heavily deforested 'Nursery Shadow' areas for restoration planners and funders.

2.3 Technology Stack

- i. Frontend: a responsive React web application serving the public portal and the nursery operator dashboard.
- ii. Web-GIS Engine: an interactive mapping library (for example Leaflet or the ArcGIS JavaScript API) rendering point, cluster, and polygon boundary layers.
- iii. Backend: a Node.js / Express REST API.
- iv. Database: PostgreSQL with the PostGIS spatial extension.
- v. Field Data Collection: ArcGIS Survey123 for nursery mapping campaigns, synchronized into the platform database.
- vi. Payments: a Ugandan mobile money API (MTN MoMo / Airtel Money) with escrow support.
- vii. Notifications: an SMS gateway for order alerts and an email service for account verification.
- viii. Satellite Data: the Global Forest Change dataset for forest-loss overlay analysis.

2.4 User Characteristics

- i. Farmers and Individual Buyers: browse, search, and order seedlings; may have varying levels of digital literacy and intermittent connectivity.
- ii. Restoration Planners and Carbon Project Developers: use the platform to locate certified suppliers at scale and to review Nursery Shadow analytics.
- iii. Nursery Operators: register their nursery, maintain live inventory and pricing, and fulfil orders.
- iv. NGOs, Government Partners (NFA, MWE), and Corporate Sponsors: list and manage free seedling campaigns.
- v. System Administrators: manage platform content, moderate nursery listings, and run Nursery Shadow analysis.

2.5 Operating Environment

Nursery Link Uganda will be hosted on a cloud server environment for scalability and maintainability. The web application must support all major desktop browsers and mobile browsers on Android 8.0 and above and iOS 13 and above, since the majority of rural users are expected to access the platform from a smartphone.

2.6 Constraints

- i. Regulatory Compliance: the system must comply with the Data Protection and Privacy Act, 2019 (Uganda) in handling user and location data.

- ii. Network Dependency: rural connectivity in Uganda can be intermittent; core browsing and cached map data should degrade gracefully under slow or interrupted connections.
- iii. Data Currency: nursery stock levels are only as accurate as the last operator update; the system cannot guarantee real-time physical stock counts.
- iv. Satellite Data Latency: the Global Forest Change dataset is updated periodically rather than in real time, so Nursery Shadow analysis reflects the most recent published forest-loss layer rather than live deforestation.

2.7 Assumptions and Dependencies

- i. Reliable, if intermittent, internet access for applicants, buyers, and nursery operators.
- ii. Continued availability of a Ugandan mobile money API for checkout and escrow.
- iii. Continued availability and periodic update of the Global Forest Change dataset and district / sub-county administrative boundary datasets.
- iv. Nursery operators' willingness to keep inventory and pricing data reasonably current.

3. Specific Requirements

3.1 External Interfaces

3.1.1 User Interfaces

- i. Public Interface: a responsive, mobile-friendly web interface for browsing, searching, and ordering.
- ii. Nursery Operator Interface: a dashboard for managing nursery profile, inventory, and pricing.
- iii. Administrator Interface: a secure console for content moderation, campaign management, and Nursery Shadow analytics.

3.1.2 Hardware Interfaces

- i. Client Devices: desktop, laptop, tablet, and mobile devices with a modern web browser and, where location features are used, GPS capability.
- ii. Server Infrastructure: scalable cloud server infrastructure hosting the application, database, and mapping services.

3.1.3 Software Interfaces

- i. Mobile Money API: for processing seedling purchase payments and escrow release.
- ii. SMS Gateway API: for automated order alerts to nursery operators.
- iii. ArcGIS Survey123 API: for importing field-collected nursery records.
- iv. Satellite Forest-Change Data Service: for the Nursery Shadow overlay analysis.

3.1.4 Communication Interfaces

- i. HTTPS: all communication between client and server is encrypted.
- ii. RESTful APIs: secure, token-authenticated REST APIs facilitate all data exchange.

3.2 Functional Requirements

ID	Requirement	Rationale	Status
FR-01	The system shall allow prospective buyers, restoration planners, and nursery operators to create accounts and sign up.	Gives every user group a persistent identity for saved searches, order history, and inventory management.	Planned
FR-02	The system shall allow registered users to securely sign in, sign out, and reset a forgotten password via a time-limited email link.	Authentication gates checkout, campaign eligibility submission, and nursery inventory editing.	Planned
FR-03	The system shall allow any visitor, authenticated or not, to browse the national nursery map and read species and news content.	Open discovery keeps the barrier to entry low for farmers with limited digital literacy.	Planned
FR-04	The system shall render an interactive Web-GIS map with three zoom tiers: national (clustered counts per administrative region), district (boundary highlight), and sub-county (tight boundary layer).	Mirrors the spatial drill-down described in the project blueprint and keeps the map legible at every scale.	Planned
FR-05	The system shall plot each nursery as a point pin at wide zoom levels and render the nursery name as a text label once the user zooms to district or sub-county level.	Prevents label clutter at the national view while keeping nurseries identifiable once the user narrows their search.	Planned
FR-06	The system shall provide a species and nursery-name search box that filters the map and the sidebar list simultaneously as the user types.	Lets a buyer searching for a specific species (for example 'Mvule') find matching nurseries without a separate lookup step.	Planned
FR-07	The system shall provide hierarchical District and Sub-County dropdown filters, where selecting a District populates the Sub-County dropdown with only that district's values.	Keeps the filtering flow structured and prevents the user from selecting an invalid district and sub-county combination.	Planned
FR-08	The system shall display a scrollable, alphabetically ordered nursery list beside the map that updates to match the current district, sub-county, or search filter.	Gives users a text-based alternative to reading pins directly off the map.	Planned
FR-09	The system shall request the user's device location on load and provide a 'Sort by Nearest' control that reorders the nursery list by calculated travel distance.	Reduces seedling transport time and transplant mortality by surfacing the closest viable supplier first.	Planned
FR-10	The system shall open a nursery detail card on pin tap or list selection, showing operator type, certification status, annual capacity, seed source, contact details, live distance, stock categories, and an itemized species inventory with unit prices.	Consolidates the information a buyer needs to decide before ordering or travelling to a nursery.	Planned
FR-11	The system shall offer a 'Follow Map to Nursery' action that converts the map into a turn-by-turn route from the user's live location to the selected nursery using road-network calculations.	Supports the self-guided pickup pathway described in the project blueprint.	Planned

ID	Requirement	Rationale	Status
FR-12	The system shall offer an 'Order & Deliver' action allowing the buyer to select species and quantities from the live inventory list.	Enables in-system purchase without a site visit.	Planned
FR-13	The system shall calculate a delivery fee automatically from the live road distance between the buyer and the nursery before checkout is confirmed.	Gives the buyer transparent, distance-based pricing consistent with local courier and transport rates.	Planned
FR-14	The system shall process seedling payments through a Ugandan mobile money API (MTN MoMo or Airtel Money) and hold funds in escrow until the buyer confirms delivery.	Protects both buyer and nursery operator from payment disputes in a cash-light rural economy.	Planned
FR-15	The system shall send an automated SMS to the nursery operator on order confirmation, containing the buyer's name, delivery coordinates, and itemized order checklist.	Reaches nursery operators who may not have constant data connectivity or app access.	Planned
FR-16	The system shall provide a Free Seedlings directory listing active giveaway campaigns, filterable by sub-county and restoration purpose, each showing a livestock meter, eligibility checklist, funder type, and pickup point.	Surfaces subsidized planting stock from NGOs, government, and corporate sponsors to equity-priority communities.	Planned
FR-17	The system shall automatically replace the standard map pin of a nursery hosting an active free seedling campaign with a distinct high-visibility gift pin.	Keeps the main nursery map and the subsidy directory in sync without manual re-tagging.	Planned
FR-18	The system shall provide a Digital Tree Library with an alphabetical index and category filters (Indigenous, Agroforestry, Exotic, Ornamental, and Medicinal).	Let's a buyer research a species before committing to an order.	Planned
FR-19	The system shall display, for each species profile, its taxonomic identity and local names, a visual growth gallery, growth pace and height-timeline milestones, canopy and root architecture notes, and preferred ecological zones.	Gives restoration planners the biological detail needed to match species to a planting site.	Planned
FR-20	The system shall provide a contextual 'Find nurseries stocking this seedling near me' action on every species profile that redirects to the Nurseries map with the species pre-filled into the search box and the map re-sorted by the user's live location.	Bridges the Library and Nurseries modules so research converts directly into a supplier search.	Planned
FR-21	The system shall underline species names inside a nursery's inventory list on the pop-up card and open a side-drawer with that species' Library entry when tapped.	Bridges the Nurseries and Library modules in the opposite direction, supporting an informed purchase decision.	Planned
FR-22	The system shall display a News and Sector Advisories feed covering weather alerts, market updates, and National Forestry Authority policy and grant announcements, filterable by category.	Keeps nursery operators and planters informed of seasonal and regulatory conditions relevant to planting decisions.	Planned

ID	Requirement	Rationale	Status
FR-23	The system shall allow authenticated nursery operators to create and update their nursery profile, inventory quantities, and unit prices from an operator dashboard.	Keeps the live stock data buyers rely on accurate without requiring administrator intervention for every update.	Planned
FR-24	The system shall allow an administrator to overlay a nursery service-area layer (5 km, 10 km, and 20 km road-network travel zones) against a satellite-derived forest-loss layer and highlight heavily deforested areas that fall outside every nursery's 20 km zone as 'Nursery Shadows'.	Implements the gap-analysis method from the underlying research concept note, directing restoration funding to the areas with the least seedling access.	Planned
FR-25	The system shall not permit a buyer to submit an Order & Deliver checkout unless species quantities, a delivery address, and a payment method have all been provided.	Prevents incomplete orders from reaching a nursery operator.	Planned

4. Non-Functional Requirements

Nursery Link Uganda must meet the following non-functional requirements to ensure it is reliable, scalable, usable, secure, and maintainable in a low-connectivity rural operating environment.

4.1 Performance Requirements

- i. NFR-1.1 System Responsiveness: complete critical operations (map load, search, and checkout) within 2-5 seconds under normal network conditions.
- ii. NFR-1.2 Map Rendering: render clustered national-level nursery pins within 3 seconds on a standard mobile connection.

4.2 Reliability and Availability

- i. NFR-2.1 System Uptime: maintain 99.5% uptime, particularly during peak planting seasons.
- ii. NFR-2.2 Data Backup: back up nursery, order, and campaign data daily with a recovery time objective of 30 minutes.

4.3 Usability and Accessibility

- i. NFR-3.1 Ease of Use: limit critical actions (finding a nursery, starting a checkout) to 3 clicks or taps from the home page.
- ii. NFR-3.2 Low-Literacy Support: rely on icons, photographs, and short labels alongside text so the interface remains usable for applicants with limited digital literacy.
- iii. NFR-3.3 Accessibility Compliance: follow WCAG 2.1 guidance for colour contrast and screen-reader compatibility where practical.

4.4 Security Requirements

- i. NFR-4.1 Data Encryption: encrypt all sensitive data, including personal information and payment references, in transit and at rest.

- ii. NFR-4.2 Access Control: enforce role-based permissions distinguishing buyers, nursery operators, and administrators.
- iii. NFR-4.3 Payment Security: never store raw mobile money credentials; all payment handling is delegated to the mobile money provider's secure API.

4.5 Scalability

- i. NFR-5.1 Horizontal Growth: support horizontal scaling of the backend API and database as the number of registered nurseries and orders grows nationally beyond the initial Mukono District pilot.

4.6 Maintainability

- i. NFR-6.1 Modular Architecture: structure the codebase into independently deployable modules (Nurseries, Free Seedlings, Library, News, Shadow Analytics).
- ii. NFR-6.2 Documentation: maintain up-to-date API and data-model documentation for future contributors.

4.7 Data Storage and Retention

- i. NFR-7.1 Retention: retain order and campaign eligibility records for a period consistent with donor and government reporting obligations.
- ii. NFR-7.2 Audit Trail: log critical actions such as inventory updates and campaign eligibility approvals.

4.8 Interoperability

- i. NFR-8.1 API Availability: expose RESTful APIs for potential integration with NFA, MWE, or donor reporting systems.
- ii. NFR-8.2 Data Export: support standard export formats (CSV, GeoJSON) for nursery and shadow-analysis data.

4.9 Environmental Sustainability

- i. NFR-9.1 Mission Alignment: the platform's core purpose is to accelerate forest restoration; system design should avoid introducing unnecessary paper-based fallback processes.
- ii. NFR-9.2 Efficient Infrastructure: optimize map tile and image delivery to reduce data costs for users on limited mobile data plans.

5. Glossary

API: Application Programming Interface - a set of rules and endpoints that allow software systems to communicate and exchange data.

Escrow: a payment arrangement in which funds are held by a neutral third party until agreed delivery conditions are met.

GIS: Geographic Information System - software used to capture, store, analyze, and present spatial data.

Nursery Shadow: a term coined in the underlying research concept note for a heavily deforested area that falls entirely outside the travel-distance service zones of every known tree nursery.

PostGIS: a spatial database extension for PostgreSQL that adds support for geographic objects.

REST API: Representational State Transfer API - a web service architecture used for communication between client and server applications.

Service Zone: a road-network-derived travel-distance boundary (5 km, 10 km, or 20 km) around a nursery, used to estimate realistic seedling transport limits.

SRS: Software Requirements Specification - a document that defines the functional and non-functional requirements of a software system.